

### Traceability Matrix

| ID | Requirements                                                                                                                                                                | Related Use Case           | Fulfilled By  | Tests                                                                                                                                              | Description                                                                                                                                                                                                      |
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| 1  | The application interface contains buttons and displays to simulate a neureset device that works with a PC and a headset                                                    | N/A                        | MainWindow.ui | Run the neureset simulator and observe the UI components included                                                                                  | The neureset system was reproduced using QtCreator's built in user interface and framework. All buttons are clickable by mouse and the application is accompanied by a console log output detailing all actions. |
| 2  | The neureset device can only be used if it has battery                                                                                                                      | Use Case #1<br>Use Case #3 | Device        | Deplete the battery by running multiple sessions. Observe that once the battery runs out, the device turns off and is unable to be turned back on. | The neureset system only operates when the battery variable in Device is above a certain amount.                                                                                                                 |
| 3  | The device has a display, green light, blue light, red light, power button, pause/resume button, stop button, up and down buttons, home button, and a select option button. | Use Case #1                | MainWindow    | Run the neureset simulator and observe the lights, the buttons, and the screen on the device                                                       | Using QtCreator's built in ui, all buttons and lights can be displayed on the main ui.                                                                                                                           |

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| 4 | The device can be connected to a headset, and a PC                        | Use Case #1 | MainWindow Device | Run the neureset simulator and observe a button to toggle headset connection on the ui. Run a session and observe that the session logs can be uploaded by clicking "view session logs" and "upload session logs"              | The Device owns a headset variable which has a variable of when the headset is connected or disconnected.                                           |
| 5 | The PC has a screen to display all the session logs                       | Use Case #4 | MainWindow        | Run the neureset simulator and notice the PC display on the right hand side. Run a session and observe that the session logs can be uploaded and displayed in a list by clicking "view session logs" and "upload session logs" | Utilizing the Qt's QListWidget, we can display and add session logs on a list and display it on the UI.                                             |
| 6 | The headset has 21 standard electrode sites                               | N/A         | Headset           | Run the neureset simulator and run a session. Observe the console log output and see that the treatments are applied to each of the 21 electrode sites                                                                         | As the program runs, it will go through the 21 electrode sites to read the frequency, send it to the device and print a message to the console log. |
| 7 | From the main menu, the user can choose to start a session, view logs, or | Use Case #1 | MainWindow        | Run the neureset simulator and notice that the main menu contains three choices to either start a new                                                                                                                          | Using Qt's QPushButton and QListWidget, we can display multiple options on the device screen and use buttons to navigate through the menu.          |

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|    | change the date and time                                                                                                        |                           |                   | session, view session logs, or change date/time                                                                                                |                                                                                                                                             |
| 8  | The user can start a new session if the device has enough battery and the headset is connected                                  | Use Case #1 / Use Case #3 | Device            | Run the neureset simulator and notice that a new session can only start when the headset connection is established and the battery is not low. | The Device checks the battery level before starting a session. If the battery level is above the required amount, a session can be started. |
| 9  | The blue light on the device turns on for the duration of the session to indicate that the headset and the device are connected | Use Case #1               | Device MainWindow | Run a new session on the neureset simulator and notice that the blue light on the device turns on for the duration of the session.             | Using QLabel, we can toggle the lights on when we start the session and turn it off when the session is not in progress.                    |
| 10 | During a session, a session timer starts and the approximate time remaining is displayed on the device screen.                  | Use Case #1               | Device MainWindow | Run a new session on the neureset simulator and notice that the session timer starts and a timer is displayed on the screen.                   | Using QLabel, we can display the remaining time by counting down from the total time needed.                                                |
| 11 | During a session, a progress bar on the display shows the current progress                                                      | Use Case #1               | Device MainWindow | Run a new session on the neureset simulator and notice the progress bar changing over the course of the session                                | Using QProgressBar, we can display the progress of the session and modify the progress throughout different stages of the session.          |
| 12 | The device                                                                                                                      | Use Case #1               | Device            | Run a new session on                                                                                                                           | The Device retrieves and calculates the                                                                                                     |

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|           | records starting baseline frequencies and amplitude of different wavelengths at every node on the headset over one minute                                         |             | Headset           | the neureset simulator and observe the console log output. Notice that the device records the baseline frequencies for every node. These can be verified on the session log after the session ends                    | correct baseline frequency from the Headset who reads the frequency.                                         |
| <b>13</b> | The starting baseline frequencies and amplitude wavelengths are recorded in the session logs and are used to calculate the dominant frequency                     | Use Case #1 | Device SessionLog | Run a new session on the neureset simulator and check the session logs for baseline frequencies and dominant frequencies after uploading it on the PC.                                                                | We store the baseline frequencies on SessionLog which stores it in startBaselines.                           |
| <b>14</b> | The treatment starts by applying a frequency equal to the dominant frequency plus an offset to each node, every 1/16th of a second for the duration of one second | Use Case #1 | Device Headset    | Run a new session on the neureset simulator and observe the console log output. Check the session logs after uploading them on the PC and notice that the different frequencies applied to each site for every round. | The Device retrieves and calculates the correct baseline frequency from the Headset who reads the frequency. |
| <b>15</b> | The offset ranges                                                                                                                                                 | Use Case #1 | Device            | Run a new session on                                                                                                                                                                                                  | We increment the offset by 5 after each                                                                      |

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|           | from 5 to 20 Hz and is increased by 5Hz for each treatment (for a total of 4 rounds of treatments)                                 |             |                | the neureset simulator and observe the frequency applied at each node in the console log output. Check the session logs after uploading them on the PC and notice that the increase of 5 Hz in frequencies applied at every round. | round, and we send the frequency to the headset to apply the correct frequencies                             |
| <b>16</b> | The green light on the device turns on for the duration of the treatment (1 sec per round)                                         | Use Case #1 | Device         | Run a new session on the neureset simulator and observe the console log output for when the device applies a treatment. Notice that the green light on the device turns green during each of the 4 rounds of treatment.            | Using QLabel, we can toggle the lights on for the duration of the treatment and off after the treatment      |
| <b>17</b> | The device records ending baseline frequencies and amplitude of different wavelengths at every node on the headset over one minute | Use Case #1 | Device Headset | Run a new session on the neureset simulator and observe the console log output. Notice that the device records the baseline frequencies for every node. These can be verified on the session log after the session ends            | The Device retrieves and calculates the correct baseline frequency from the Headset who reads the frequency. |

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| <b>18</b> | The ending baseline frequencies and amplitude wavelengths are recorded in the session logs after the session is over                                                                                                       | Use Case #1 | Device<br>SessionLog | Run a new session on the neureset simulator and check the session logs for baseline frequencies and dominant frequencies after uploading it on the PC.       | We store the baseline frequencies on SessionLog which stores it in endBaselines.                                                                |
| <b>19</b> | After the session has been completed and the ending baseline frequencies and amplitude wavelengths have been recorded in the session logs, the session log is saved to the device and is ready to be transferred to the PC | Use Case #1 | Device<br>SessionLog | Run a new session on the neureset simulator and check the session logs on the device. Try uploading the session logs onto the PC and view the data there.    | The session log that was saved is stored in a vector of session logs which saves can save multiple session logs.                                |
| <b>20</b> | A session cannot start if the headset is not connected to the device and to the patient's head                                                                                                                             | Use Case #1 | Device               | Run a new session on the neureset simulator after toggling the headset connection and notice that a new session cannot start unless the headset is connected | A variable in Device checks if the connection from the Headset and Device is made. If the connection is not there, a session cannot be started. |
| <b>21</b> | If electrodes lose contact with the                                                                                                                                                                                        | Use Case #1 | Device               | Run a new session on the neureset simulator                                                                                                                  | A variable in Device checks if the headset is in contact with the patient's                                                                     |

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|           | patient's head, the current session will be paused                                                                                                        |             |                      | and toggle the headset connection. Notice that the current session is paused.                                                                                                                                                                            | head. If it is not, the current session will be paused.                                                                                                         |
| <b>22</b> | The user can pause the current session by pressing the pause/resume button on the device.                                                                 | Use Case #1 | Device               | Run a new session on the neureset simulator and click the pause/resume button. Notice that the session progress will toggle as you resume or pause the session.                                                                                          | Using a QPushButton, we can call the function that pauses the session by pressing the pause button. If the button is pressed again, it will resume the session. |
| <b>23</b> | When the current session is paused, the red light on the device will start flashing, the device will start beeping, and a timer for 5 minutes will start. | Use Case #1 | Device               | Run a new session on the neureset simulator and pause the session. Notice that the red light is flashing on the UI and a timer for 5 minutes has started on the console log output.                                                                      | Using QLabel, we can toggle the lights on if the session is paused.                                                                                             |
| <b>24</b> | If the 5-minute timer times out before re-establishing contact and/or resuming the session, the device will turn off and erase the session log.           | Use Case #1 | Device<br>SessionLog | Run a new session on the neureset simulator and pause the session/toggle the headset connection. Wait until the timer times out and notice that the device turns off. Turn the device back on and notice that the session does not appear in the session | When the timer times out, the session function will be returned and nothing is saved.                                                                           |

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|           |                                                                                                                                                          |             |                  | logs                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                    |
| <b>25</b> | If contact is re-established and/or the current session is resumed before the 5-minute timer times out, the current session will resume                  | Use Case #1 | Device           | Run a new session on the neureset simulator and pause the session/toggle the headset connection. Resume the session/toggle the headset connection again before the timer times out and notice that the session resumes                                                                                                                                                    | The Device allows the session to resume if the variable for the connection is re-established, and the session will start from where it was paused. |
| <b>26</b> | When the current session is resumed, the red light will stop flashing, the device will stop beeping, and the session will resume from where it left off. | Use Case #1 | Device           | Run a new session on the neureset simulator and pause the session/toggle the headset connection. Resume the session/toggle the headset connection again before the timer times out and notice that the red light on the device stops flashing, a message saying the device stops beeping is outputted to the console log, and the session resumes from where it left off. | The QLabel representing the lights will be toggled off and the light will stop flashing, and the device will stop beeping.                         |
| <b>27</b> | The user can update the current                                                                                                                          | Use Case #2 | Device QDateTime | Run the neureset simulator and click                                                                                                                                                                                                                                                                                                                                      | The device has a QDateTime that can be updated to the date specified by the user.                                                                  |



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|           | date and time on the device                                                                                                                               |                              |                      | “Time and date”. Notice that when the time and/or date is changed and the OK button is pressed, the change is reflected in subsequent session logs. This can be verified by viewing session logs on the device or on the PC after uploading |                                                                                                                                                                                                                                                                                                 |
| <b>28</b> | At any moment, the device detects low battery                                                                                                             | Use Case #3                  | Device               | Run the neureset simulator and start a new session until the battery low message pops up on the screen.                                                                                                                                     | A timer is used to send a signal at specified intervals that update the battery usage.                                                                                                                                                                                                          |
| <b>29</b> | When the device detects low battery, a message is displayed on the device for 5 seconds and the low battery icon is displayed in the corner of the screen | Use Case #3                  | Device<br>MainWindow | Run the neureset simulator and start a new session until the battery low message pops up on the screen. Notice that this message is displayed on the screen for 5 seconds and the battery icon is changed to the low battery icon.          | Using the member variable in the Device class, we can track the battery percentage. Each time that the battery is decreased, it checks if it is under a threshold of 20%. If it is, it will display a low battery message on the display to notify the user and a new session cannot be started |
| <b>30</b> | If the device detects low battery during a session, it finishes the session and is unable to start a                                                      | Use Case #1 /<br>Use Case #3 | Device               | Run the neureset simulator and start a new session until the battery low message pops up on the screen in the middle of the                                                                                                                 | Using the member variable in the Device class, we can track the battery percentage. Each time that the battery is decreased, it checks if it is under a threshold of 20%. If it is, it will finish the current session, display a low battery                                                   |

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|           | new session until the battery is replaced.                                                                                                           |                           |                      | session. Notice that the session finishes and the logs are saved properly but a new session with low battery cannot be started.                                                                                                                                                                   | message on the display to notify the user and a new session cannot be started                                                                                                                                                                     |
| <b>31</b> | If the device detects low battery during a session, the user can pause the session to change the battery and resume the session after battery change | Use Case #1 / Use Case #3 | Device               | Run the neureset simulator and start a new session until the battery low message pops up on the screen in the middle of a session. Press the pause button and the change battery button. Turn on the device again and resume the session. Notice that the session resumes from where it left off. | Using the signals and slots framework of Qt, the user pressing the pause QPushButton will pause the session and pressing the replace QPushButton will automatically replenish the battery to 100%                                                 |
| <b>32</b> | If the device battery is completely depleted, the device turns off automatically.                                                                    | Use Case #3               | Device               | Run the neureset simulator and start a new session until the battery low message pops up on the screen in the middle of a session. Wait until the battery is completely depleted and notice that the device turns off                                                                             | Using the member variable in the Device class, we can track the battery percentage. Each time that the battery is decreased, it checks if it is under a threshold of 20% and equal to 0%. Once it reaches 0%, the device turns off automatically. |
| <b>33</b> | The user can press "session log                                                                                                                      | Use Case #4               | Device<br>SessionLog | Run the neureset simulator and start a                                                                                                                                                                                                                                                            | Utilizing the Qt's QListWidget, we can display and add session logs on a list                                                                                                                                                                     |

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|           | history” to view a list of all previous session logs displaying the date and time for each session log                                                                       |             |                            | new session. Once it has finished, navigate to the home menu and click on “Session Logs”. Observe that the session logs are present.                                              | and display it on the UI.                                                                                                                                       |
| <b>34</b> | The user can select the “upload session logs” to upload all the session logs from the device to the PC, if the device is connected to a PC                                   | Use Case #4 | Device<br>PC<br>SessionLog | After running a session, navigate to “Session Logs” and click the “Upload Logs” button. Observe that the logs are transferred to the PC, where they can be viewed in more detail. | The device and PC have separate stores for their logs. Uploading from the device to the PC involves simply copying the information from one store to the other. |
| <b>35</b> | Once session logs are uploaded onto the PC, the PC stores and displays all uploaded session logs                                                                             | Use Case #4 | PC<br>SessionLog           | After running a session, navigate to the ‘Session Log’ screen of the device, click on ‘Upload logs’, the logs should now be visible on the PC view.                               | The PC holds the information of all its uploaded logs in memory. The display gathers that information to show an exhaustive list of the logs displayed.         |
| <b>36</b> | The user can click into a session log stored on the PC which will display the date, start time, end time, baseline dominant frequencies per site before treatment, treatment | Use Case #4 | PC<br>SessionLog           | After uploading a session log, double clicking on the corresponding list item on the PC view, the details of the log should be displayed on the same screen.                      | By leveraging a signal from the list widget in the session history, we can find and display the information from the log stored in the PC.                      |

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|  | frequencies applied over the 4 rounds, and baseline dominant frequencies per site after treatment |  |  |  |  |
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